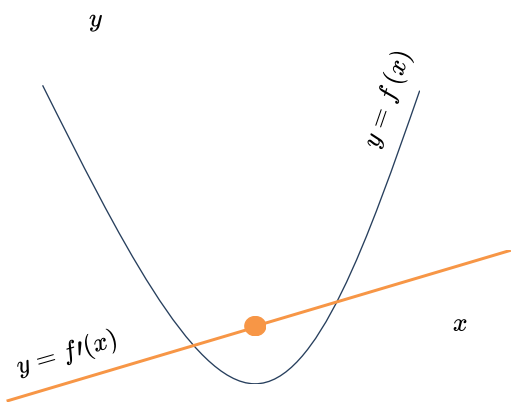
1.33 Differentiation: Graphs of Gradient Functions

Chapter 12 · Original slide 64 · Sketching Gradient Functions



The new A Level specification specifically mentions being able to sketch $y = f\prime(x)$.

If you know the function $f\prime(x)$ explicitly (e.g. because you differentiated $y = f(x)$), you can use your knowledge of sketching straight line/ quadratic/cubic graphs.

But in other cases you won't be given the function explicitly, but just the sketch.

The gradient of $y = f(x)$ is negative, but increasing.

The gradient at the turning point is 0.

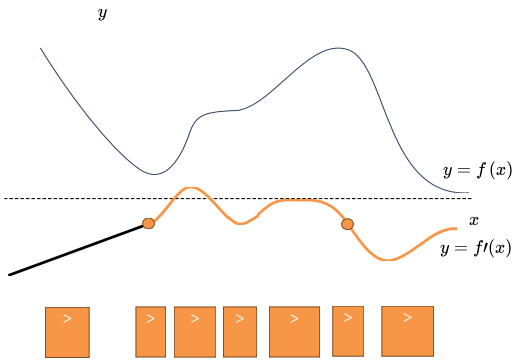
The gradient of $y = f(x)$ is positive, and increasing.

Click to Sketch $y = f\prime(x)$

Click to Sketch $y = f\prime(x)$

Click to Sketch $y = f\prime(x)$

Chapter 12 · Original slide 65 · A Harder One



Tip: Mentally describe the gradient of each section of the curve, starting your mental sentence with "The gradient is…" and using terms like "positive", "but increasing", "but not changing", etc.

Gradient is negative, but increasing.

Gradient is positive: initially increases but then decreases.

Gradient is positive: initially increases but then decreases.

Gradient is 0.

Gradient is 0, but positive before and after.

Gradient is 0.

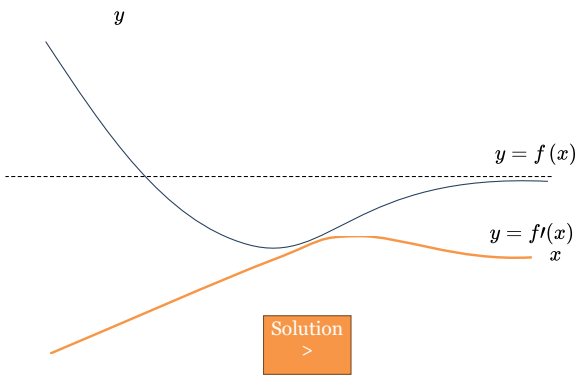
Gradient is negative. Initially decreases but then tends towards 0.

Tip: The gradient is momentarily not changing so the gradient of the gradient is 0. We get a turning point in $y = f'(x)$ whenever the curve has a point of inflection.

Again, on $y = f(x)$ this is a point of inflection so the gradient function has a turning point. It is also a

1.33 Differentiation: Graphs of Gradient Functions

Chapter 12 · Original slide 66 · Test Your Understanding



Solution >

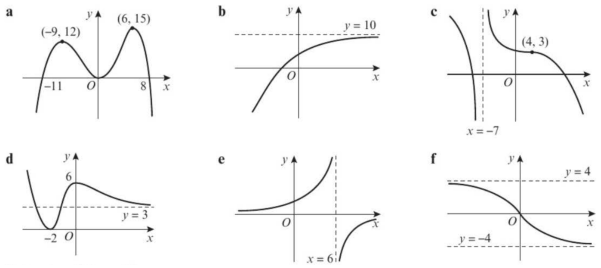
Gradient is negative, but increasing.

Gradient is positive; initially increases but then decreases.

Gradient is positive, but tending towards 0.

Gradient is 0.

Chapter 12 · Original slide 67 · Exercise 12J – Pearson Pure Y1 page 278



1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.

Solutions here

1.33 Differentiation: Graphs of Gradient Functions

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2 $\mathrm{f}(x) = (x+1)(x-4)^2$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On a separate set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Show that $\mathrm{f}'(x) = (x-4)(3x-2)$.
d Use the derivative to determine the exact coordinates of the points where the gradient function cuts the coordinate axes.

Chapter 12 · Original slide 69 · Extension

1
[MAT 2015 1B]
$f(x) = (x+a)^n$
where a is a real number and n is a positive whole number, and $n \geq 2$. If $y = f(x)$ and $y = f\prime(x)$ are plotted on the same axes, the number of intersections between $f(x)$ and $f'(x)$ will:

A) always be odd

B) always be even

C) depend on a but not n

D) depend on n but not a

E) depend on both a and n.

Recall that:

Odd function: $f(-x) = -f(x)$

Even function: $f(-x) = f(x)$